# Ghost Target Generation via Post-Processing DOA Spoofing: Hardware Validation on USRP SDR

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***Abstract*—Direction-of-Arrival (DOA) estimation is a key component of array-based sensing and radar-oriented perception systems, and its reliability is particularly important in safety-critical applications. Recent work has shown that DOA estimation can be intentionally manipulated through physical-layer spoofing, but existing evaluations remain largely simulation-based. In this work, a hardware-based study of a post-processing variant of DOA spoofing is presented using commercial USRP software-defined radios. Because the available setup includes only a single transmit antenna, the original multi-antenna over-the-air attack is not replicated; instead, the underlying spoofing transformation is applied directly to recorded IQ samples in order to assess its behavior on real RF measurements. Experimental results show that the MUSIC estimator can be redirected across the interval $[-60^\circ, +60^\circ]$ with a 100% success rate under the adopted test conditions. The study also highlights practical non-idealities that emerge on real hardware, including phase calibration offsets, multipath, and angular aliasing, and provides a reproducible baseline for future investigations toward a full physical implementation.**

***Index Terms*—Direction-of-Arrival estimation, DOA spoofing, physical-layer attacks, USRP, software-defined radio, MUSIC algorithm, autonomous driving security.**

## I. Introduction

Direction-of-Arrival (DOA) estimation is a fundamental task in array signal processing and plays a central role in radar-based perception systems, wireless localization, and other sensing applications based on antenna arrays. By estimating the angular direction from which a signal arrives, DOA techniques support target localization and higher-level interpretation of the surrounding environment. In safety-critical scenarios, the reliability of this estimate becomes especially important, since erroneous angular information may propagate to subsequent detection and decision-making stages.

In recent years, increasing attention has been devoted not only to the performance of DOA estimation algorithms, but also to their security implications. If the estimated direction of an incoming signal can be intentionally manipulated, a receiver may construct a distorted representation of the environment, potentially leading to incorrect localization of real targets or the perception of non-existent ones. This makes DOA estimation a relevant attack surface in radar-oriented and array-based sensing systems.

Among the recent contributions in this area, Xu et al. [1] introduced a DOA spoofing attack capable of steering the estimated angle of arrival toward a desired ghost direction by controlling the phase relationships observed at a multi-antenna receiver. Their work established an important proof of concept, but its evaluation was conducted in simulation and did not include a validation on real hardware. As a result, it remains unclear to what extent the mathematical core of the spoofing transformation holds under practical RF conditions affected by calibration errors, multipath, and hardware non-idealities.

**Our Contribution.** This paper addresses that question in a constrained experimental setting based on commercial USRP software-defined radios. Since the available transmitter configuration does not satisfy the assumptions required for the original multi-antenna over-the-air attack, the goal is not to reproduce the complete physical-layer threat model. Instead, the work investigates a post-processing variant in which the spoofing transformation is applied directly to recorded IQ samples, making it possible to evaluate its behavior on real measurements while remaining compatible with the available hardware.

The contribution of the paper is therefore twofold. First, it provides a hardware-based validation of the mathematical core of the DOA spoofing transformation on real RF data. Second, it helps clarify the distinction between algebraic validation in a constrained setup and full physical realization of the original attack, thereby establishing a reproducible baseline for future work on synchronized multi-antenna implementations.

The remainder of the paper is organized as follows. Section II reviews the main related work. Section III introduces the reference model and the theoretical elements used in the analysis. Section IV defines the threat model and presents the post-processing variant considered in this work. Section V describes the experimental setup, Section VI reports the obtained results, Section VII discusses their interpretation and limitations, Section VIII outlines possible future developments, and Section IX concludes the paper.

## II. Related Work

**Direction-of-arrival (DOA) estimation** has been studied for decades, and classical subspace methods such as MUSIC [2] and ESPRIT [3] remain among the most widely used techniques for high-resolution angle estimation. More recently, DOA estimation has been applied in wireless localization, room occupancy sensing, and physical-layer authentication, showing its relevance well beyond traditional radar processing.

**Security of DOA systems.** Most prior work has focused on leveraging DOA for defensive purposes rather than attacking it. Examples include beamforming-based protection and physical-layer authentication, while only a limited number of studies have considered adversarial manipulation of angle estimation. Abdelaziz et al. [4] investigated vulnerability to hostile jamming, but only under a fixed interference angle. Yang et al. [5] proposed adversarial attacks against deep learning-based DOA estimators, showing that data-driven methods can also be misled. In the GNSS spoofing literature, DOA-based detection has also been studied, but the fixed orbital positions of satellites make such attacks structurally different from general radar settings.

The security implications of DOA estimation have received growing attention. A compromised DOA estimator could cause an autonomous vehicle to detect non-existent obstacles, brake unexpectedly, or fail to detect real pedestrians — scenarios with potentially fatal consequences. To the best of our knowledge, Xu et al. [1] were the first to propose a variable-angle physical-layer DOA spoofing attack targeting general radar receivers. Their work introduced a new attack surface for array-based perception systems and showed, in simulation, that the receiver's estimated angle can be manipulated toward a desired ghost direction. At the same time, their evaluation was entirely simulation-based and did not include a hardware validation of the attack model.

**SDR-based security research.** Software-defined radios have already been widely used in the security literature to demonstrate physical-layer attacks in practice, including Wi-Fi frame injection and GPS spoofing. The present work follows this line of research and uses USRP hardware to study how the mathematical core of the DOA spoofing transformation behaves on real RF measurements.

## III. System Model

### A. Signal Model

We adopt the signal model introduced by Xu et al. [1]. The receiver is equipped with $N$ omnidirectional antennas arranged in a uniform linear array (ULA) with inter-element spacing $d_r = \lambda/2$, where $\lambda$ is the carrier wavelength. For a signal arriving from angle $\xi$, the phase difference between adjacent antenna elements is

$$\Delta\phi = 2\pi \frac{d_r}{\lambda} \sin(\xi) = \pi \sin(\xi). \tag{1}$$

The corresponding steering vector for angle $\xi$ with N antennas is

$$\mathbf{a}(\xi) = \left[1, e^{j\pi \sin(\xi)}, \ldots, e^{j(N-1)\pi \sin(\xi)}\right]^T. \tag{2}$$

Let $\mathbf{Y} \in \mathbb{C}^{T\times N}$ denote the received signal matrix over $T$ time samples. Under a single-source model at angle $\theta$, the received signal can be written as

$$\mathbf{Y} = \mathbf{s}(t)\mathbf{a}(\theta)^T + \mathbf{N}, \tag{3}$$

where $\mathbf{s}(t) \in \mathbb{C}^{T\times 1}$ is the source waveform and $\mathbf{N}$ is additive noise.
In our hardware setup, we consider a two-element array ($N = 2$) operating at carrier frequency $f_c = 2.4$ GHz, which gives $\lambda \approx 12.5$ cm and $d_r \approx 6.25$ cm. The single transmitter ($M = 1$) used in the experiment is a USRP B200 operating at the same carrier frequency.

### B. MUSIC Estimator

Given the received matrix $\mathbf{Y}$, the MUSIC algorithm computes the spatial covariance matrix

$$\mathbf{R} = \frac{\mathbf{Y}^H\mathbf{Y}}{T},$$

and decomposes it into signal and noise subspaces. Let $\mathbf{E}_n$ denote the matrix of noise eigenvectors. The MUSIC pseudo-spectrum is then defined as

$$P_{\text{MUSIC}}(\xi) = \frac{1}{\mathbf{a}(\xi)^H\mathbf{E}_n\mathbf{E}_n^H\mathbf{a}(\xi)}. \tag{4}$$

The estimated direction of arrival is obtained as the angle that maximizes this pseudo-spectrum. In practice, MUSIC provides a high-resolution angular estimate and is widely used in array processing because of its ability to resolve closely spaced sources under suitable signal conditions.

## IV. Threat Model and Post-Processing DOA Spoofing Methodology

### A. Threat Model (Xu et al.)

The DOA spoofing attack proposed by Xu et al. [1] is formulated for a physical-layer setting in which an attacker controls multiple synchronized transmit antennas and injects carefully designed signals over the air. Under these assumptions, the received wavefront can be manipulated so that a multi-antenna radar estimates a desired ghost direction instead of the true one. The core of the method relies on the availability of $M \geq N$ transmit antennas, full-rank coverage of the receiver signal space, and coherent transmission from all attacker antennas.

These assumptions are not merely implementation details. They are essential to the original attack model because the transformation is meant to act on the physical superposition of the transmitted waves before they reach the radar receiver. In other words, the original contribution of Xu et al. [1] is tied to a real over-the-air injection scenario, not to an offline manipulation of already acquired data.

Specifically, in the original formulation [1], an attacker with $M$ transmit antennas constructs a weight vector $\mathbf{w} \in \mathbb{C}^M$ such that the superposition of their transmitted signals, as received by the $N$-antenna radar, mimics a signal arriving from a desired ghost angle $\phi$. The transformation matrix is:

$$\mathbf{H} = \mathbf{A}(\theta)^H \left(\mathbf{A}(\theta)\mathbf{A}(\theta)^H\right)^{-1} \mathbf{A}(\phi) \tag{5}$$

where $\mathbf{A}(\theta)$ and $\mathbf{A}(\phi)$ are the steering matrices for the attacker's true angles and the desired ghost angles, respectively. This requires $M \geq N$ for full-rank coverage of the receiver's signal space, and synchronized transmission from all $M$ antennas.

### B. Why the Original Setup Could Not Be Replicated

The experimental setup considered in this work does not allow a faithful replication of those assumptions. The available transmitter is a single USRP B200, so the condition $M \geq N$ required by the original attack is not satisfied. As a consequence, the over-the-air spoofing mechanism cannot be implemented in the same way as in the reference paper, because there is no multi-antenna transmission array to synthesize the intended field at the receiver.

In addition, real hardware introduces further complications that the original simulation-based evaluation does not account for, such as independent oscillator phase offsets, indoor multipath, and imperfect calibration. For this reason, the goal of the present work is not to reproduce the full physical attack, but to isolate and validate the mathematical transformation underlying it in a controlled experimental setting.

### C. Post-Processing Variant (Our Work)

To adapt the original idea to the available hardware, the spoofing transformation is applied directly to recorded IQ samples instead of being realized through over-the-air transmission. This choice makes it possible to test the effect of the transformation on real RF measurements while avoiding the need for synchronized multi-antenna transmission. The result is a post-processing variant that preserves the algebraic structure of the original method but operates on already captured data. With $M = 1$, the over-the-air attack is infeasible; instead, the same transformation is applied directly to the recorded IQ matrix $Y$. For the case $M = 1$, $N = 2$, $L = 1$ (one ghost target), both $A(\theta)$ and $A(\phi)$ reduce to column vectors $a(\theta)$ and $a(\phi)$, and Equation (5) simplifies to a scalar projection.

The procedure is intentionally simple: first, the source contribution is extracted from the measured signal by projecting the recorded matrix onto the steering vector associated with the baseline direction; then, the extracted source component is reconstructed using the steering vector corresponding to the desired ghost angle. Feeding the resulting modified observation into MUSIC causes the estimator to converge to the target ghost direction.

We apply the transformation in two steps. First, we extract the source signal by projecting $Y$ onto the real steering vector:

$$\hat{s}(t) = Y \, \frac{a(\theta)^*}{\|a(\theta)\|^2} \tag{6}$$

Then, we reconstruct the ghost observation:

$$Y' = \hat{s}(t)\, a(\phi)^T \tag{7}$$

Feeding $Y'$ to the MUSIC estimator yields $\hat{\phi}$ instead of $\hat{\theta}$, effectively redirecting the radar's perception to the ghost direction.

The MATLAB implementation of this transformation is:

```
% Steering vectors (Eq. 2)
a_theta = [1; exp(1j * pi * sind(theta_real))];
a_phi   = [1; exp(1j * pi * sind(phi_ghost))];

% Step 1: extract source signal (Eq. 6)
s_hat = Y * conj(a_theta) / (a_theta' * a_theta);

% Step 2: reconstruct ghost observation (Eq. 7)
Y_prime = s_hat * a_phi.';
```

Fig. 1. Post-processing spoofing transformation.

### D. Obtaining $\theta$ from Data

Since our transmitter placement was subject to calibration offsets (see Section V), we do not use the nominal angle as $\theta$. Instead, we obtain $\theta = \hat{\theta}_{\text{baseline}}$ directly from the MUSIC estimate on the unmodified $Y$. This ensures that the transformation is grounded in the system's actual measured state.

### *E. Relation to the Reference Paper*

This variant should be interpreted as a validation of the mathematical core of the attack rather than as a full implementation of the physical-layer threat described in the reference paper. The original work remains the reference for the complete multi-antenna spoofing scenario, while the present contribution shows that the same transformation can be meaningfully applied to real hardware data even under a significantly constrained setup. This distinction is central to the paper and will be reflected throughout the experimental discussion.

### *F. Comparison with Original Attack*

Table I summarizes the key differences between the original attack and our post-processing variant.

TABLE I
COMPARISON: ORIGINAL ATTACK VS THE POST-PROCESSING VARIANT.

| Feature | Original Attack | Our Work |
|---|---|---|
| Transmit antennas M | $\geq N$ | 1 |
| Signal injection | Over-the-air | Post-processing |
| Real hardware | No (simulation) | Yes (USRP) |
| Synchronization required | Yes | No |
| Real-world impairments | Not modeled | Measured |

## V. EXPERIMENTAL SETUP

### *A. Hardware Configuration*

The receiver is a **USRP X310** equipped with two independent UBX daughterboards, each connected to one antenna[1]. Because the two daughterboards use independent local oscillators (LOs), the receiver exhibits a systematic phase offset between channels and therefore requires software-based phase calibration at startup. The transmitter is a **USRP B200** connected via USB to a separate laptop. Both radios operate at $f_c = 2.4$ GHz, and the transmitter generates a complex tone with a $f_{\text{tone}} = 100$ kHz offset from the carrier. On the receiver side, the signal is sampled at $f_s = 400$ kHz, and the two receive antennas are connected to the RX2 ports of their respective daughterboards and spaced by approximately $\lambda/2 \approx 6.25$ cm, forming a two-element ULA. Figure 2 and Figure 3 show respectively the deployed transmitter and receiver.


Fig. 2. USRP B200 transmitter with a single antenna.

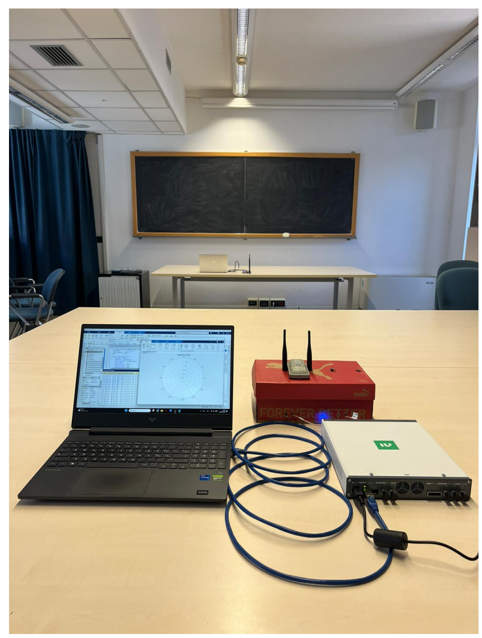

Fig. 3. USRP X310 receiver with two UBX daughterboards driving the two-element $\lambda/2$ ULA (the two antennas), connected via Ethernet to the host running the MUSIC estimator.

### *B. Phase Calibration*

Since the two UBX daughterboards are driven by independent oscillators, a systematic phase offset $\phi_{\text{cal}}$ exists between the two received channels. At the beginning of each session, with the transmitter positioned at broadside (0°), this offset is estimated and compensated using the average phase relation between the two channels:

$$\phi_{\text{cal}} = \angle \left( \mathbb{E} \left[ y_1(t) \cdot y_2(t) \right] \right) \tag{8}$$

The correction is then applied to the second channel as $y_2^f(t) = y_2(t)\, e^{-j\phi_{\text{cal}}}$ before any DOA estimation is performed.

[1]The implementation is inspired by the MATLAB USRP Radio example for DOA estimation [6], adapted for a two-UBX configuration with independent local oscillators requiring software-based phase calibration.

A limitation of this approach is that any misalignment of the transmitter from the true broadside direction during calibration would bias all subsequent angle estimates. In our setup, the broadside dataset is correctly estimated near 0°, indicating the calibration was accurate; nonetheless, we use MUSIC-estimated angles (not nominal ones) as ground truth throughout our evaluation.

### C. Data Collection

IQ data are collected with the transmitter placed manually in three approximate positions: broadside (≈ 0°), and two wide-angles (≈ ±40°). For the wide-angle positions, repeated acquisitions are performed to capture run-to-run variability. Each acquisition stores 100 frames of 4096 samples, for a total of 409,600 samples, corresponding to roughly 1 second of data. The broadside dataset is estimated near 0, confirming the phase calibration, whereas the wide-angle datasets show a consistent left–right sign inversion: the MUSIC estimate has opposite sign to the nominal position. Because the transmitter was positioned manually, we report the MUSIC-estimated DOA, which is the quantity used as ground truth $\theta$ for the spoofing transformation. Table II summarizes the collected datasets.

TABLE II
Collected IQ datasets (409,600 samples each, ≈ 1 s). Transmitter positions were set manually and are approximate.

| Dataset | Approx. TX azimuth | MUSIC-estimated DOA |
|---|---|---|
| D1 | ≈ 0 (broadside) | +1.5 |
| D2 | ≈ +40 | −11.5 to −40 |
| D3 | ≈ −40 | +47.5 to +49 |

### D. Dataset Summary

The broadside dataset is estimated close to 0°, confirming that the phase calibration is working. The two wide-angle datasets show a consistent left-right sign inversion, meaning that the MUSIC estimate often has the opposite sign with respect to the nominal transmitter position. This behavior is consistent with the physical ordering of the two antennas and does not affect the spoofing procedure, since the transformation is based on the measured baseline angle.

### E. Public Implementation

All code used for the experiment is publicly available in the repository referenced by the paper at [7]. This includes the calibration routine, the MUSIC-based angle estimation, and the post-processing spoofing transformation applied to the recorded IQ samples.

## VI. Experimental Results

### A. Baseline DOA Estimation

The first step of the evaluation is the estimation of the baseline direction from the unmodified IQ recordings. For each dataset, the MUSIC algorithm is applied before any spoofing transformation in order to obtain the baseline angle $\theta = \hat{\theta}_{\text{baseline}}$ used in the subsequent processing. In the broadside acquisition, the estimated direction is close to 0°, which is consistent with the calibration procedure. For the wide-angle datasets, the estimated peaks appear at angles whose sign is inverted with respect to the nominal manual placement, a behavior that is consistent across acquisitions and reflects the physical ordering of the two receive antennas rather than a failure of the estimator.

Figure 4 shows a representative baseline spectrum for one of the wide-angle datasets (transmitter at ≈ +40°), with a clear main peak at approximately −40°. At the same time, the spectrum is characterized by a relatively broad main lobe, which is expected for a two-element array ($N = 2$ antenna elements), and by the presence of a secondary lobe attributable to indoor multipath. These observations are important because they show that the measurements are obtained under realistic non-ideal conditions rather than under a perfectly clean simulated scenario.

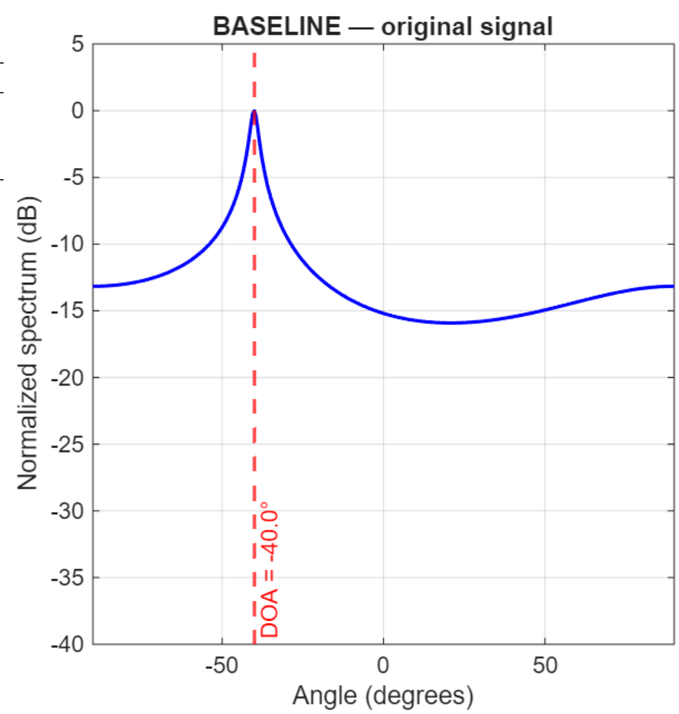


Fig. 4. MUSIC baseline spectrum for a wide-angle dataset (transmitter at ≈ +40°). The peak at −40° is the MUSIC-estimated DOA; its sign is inverted relative to the nominal position, reflecting the physical ordering of the two antennas. The broad mainlobe is characteristic of a two-element array.

### B. Single Ghost-Target Spoofing

After establishing the baseline direction, the post-processing spoofing transformation is applied to redirect the estimated arrival angle toward a selected ghost direction. In the representative example considered here, the baseline angle is approximately $\theta = -40^\circ$, and the transformation is used to force the estimator toward a ghost target at $\phi_{\text{ghost}} = +50^\circ$. This corresponds to a total angular displacement of $90^\circ$ with sign reversal.

Figure 4 shows the baseline peak at $-40^\circ$; Figure 5 shows the resulting MUSIC spectrum after the transformation is applied: the dominant peak moves from the original baseline direction $-40^\circ$ to the desired ghost angle $+50^\circ$. In this test case, the redirected estimate matches the target angle at exactly $+50^\circ$, yielding zero angular error. This result confirms that, within the considered setup, the transformation is able to alter the angular perception of the receiver in a fully controlled way on real recorded RF data.

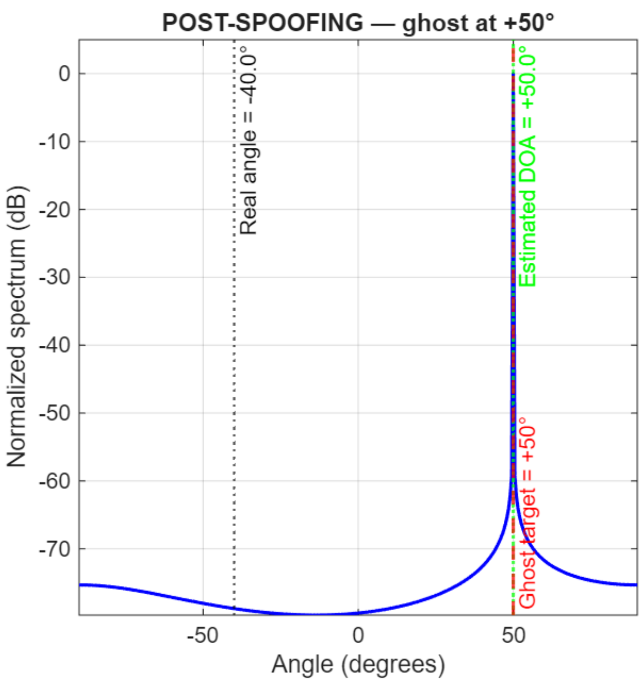


Fig. 5. MUSIC spectra after applying the post processing spoofing transformation. The transmitter is at $\theta = -40^\circ$; the ghost target is placed at $\phi_{\text{ghost}} = +50^\circ$, a displacement of $90^\circ$ with sign change. Error = $0.00^\circ$.

### C. Ghost-Angle Sweep

To evaluate whether the observed behavior is limited to a single example or generalizes across different targets, the experiment is extended to a sweep of ghost angles in the interval $[-60^\circ, +60^\circ]$ with a step of $5^\circ$, yielding a test set $\Phi_{\text{test}}$ of 25 target angles. For each $\phi \in \Phi_{\text{test}}$, the absolute angular error $|\hat{\phi} - \phi|$ between the estimated direction and the desired ghost angle is recorded. We define the success rate as

$$\text{SR} = \frac{\left|\{\phi \in \Phi_{\text{test}} : |\hat{\phi} - \phi| < \varepsilon\}\right|}{|\Phi_{\text{test}}|}, \tag{9}$$

where $\varepsilon = 1^\circ$ is the angular error threshold. Figure 6 shows the estimated DOA versus the target angle across the full sweep. The estimated curve perfectly overlaps the ideal $y = x$ line for all 25 tested angles, meaning the error is practically zero throughout the range and SR = 100% according to (9).

This does not mean that the full physical attack has been demonstrated under all practical constraints, but it does show that the post-processing transformation remains internally consistent and exact when exercised on real hardware measurements.

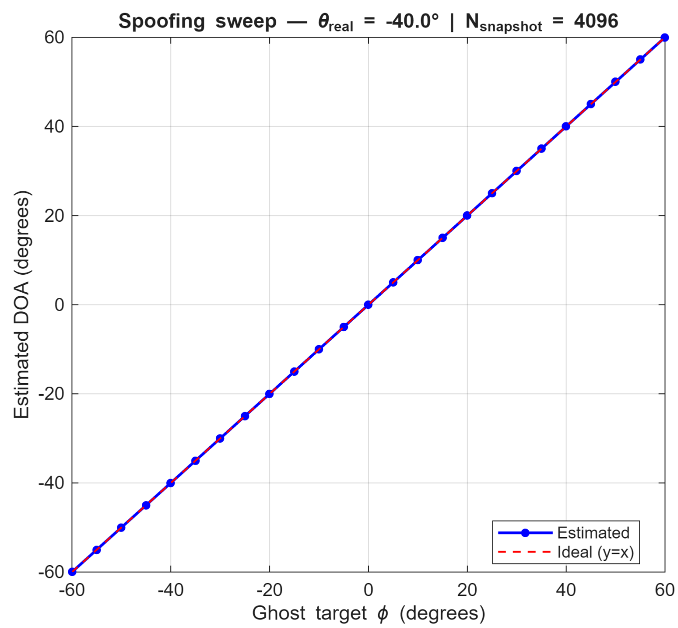


Fig. 6. Ghost angle sweep from $-60^\circ$ to $+60^\circ$ in $5^\circ$ steps (25 target angles). The estimated DOA perfectly overlaps the ideal $y = x$ line for every tested angle, yielding SR = 100% as defined in (9).

### D. Angular Aliasing Outside the Operating Range ($\pm 90^\circ$)

An additional test is performed at $\phi_{\text{ghost}} = +100^\circ$ to explore the behavior of the method outside the intended operational region. In this case, the estimator returns approximately $+80^\circ$, resulting in an error of about $-20^\circ$. This is not interpreted as a failure of the spoofing transformation itself, but rather as a consequence of angular aliasing in the two-element ULA geometry.

The reason is that angles producing the same phase difference cannot be distinguished by the array, and in particular $\sin(100^\circ) = \sin(80^\circ)$. The observed ambiguity is therefore a structural limitation of the array model and confirms that the effective operating region is bounded by the angular identifiability of the receiver by $\pm 90°$, consistent with the $[-60°, +60°]$ range recommended by Xu et al. [1]. This behavior is illustrated in Figure 7. In this sense, the out-of-range test is useful because it clarifies the limits within which the proposed variant can be meaningfully evaluated.

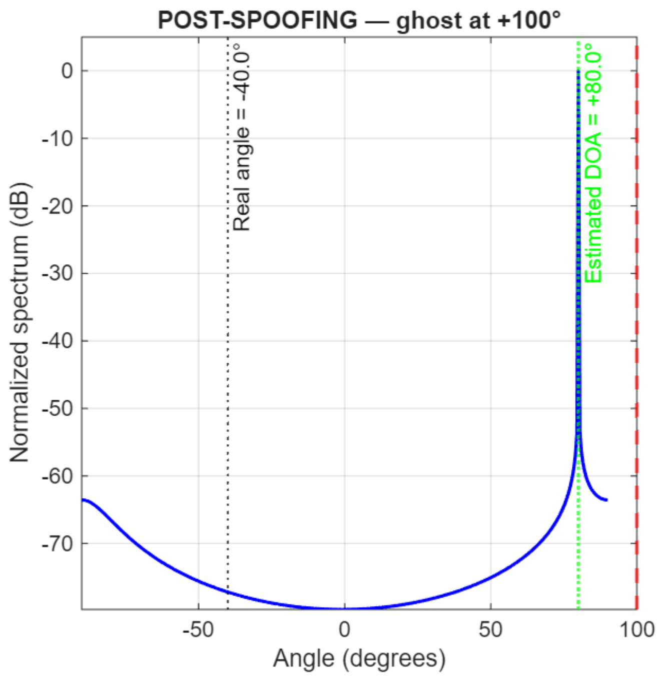


Fig. 7. Angular aliasing at $\phi_{\text{ghost}} = +100°$ (in red): MUSIC returns $+80°$ (error: $-20°$) due to the symmetry $\sin(100°) = \sin(80°)$. This is a fundamental geometric limit of the ULA, not a failure of the transformation.

### E. Interpretation of the Results

Taken together, the results indicate that the considered transformation behaves as expected on real IQ acquisitions obtained with the USRP-based setup. The baseline estimation, the controlled redirection of a single target, and the complete sweep over the $[-60°, +60°]$ interval all support the same conclusion: the mathematical core of the spoofing approach remains valid when applied to experimentally measured signals. At the same time, the measurements also reveal practical effects such as multipath, calibration sensitivity, and angular ambiguity, which motivate a careful distinction between this proof of concept and a complete over-the-air realization of the original attack model.

## VII. Discussion

### A. Interpretation of the 100% Success Rate

At first sight, the reported 100% success rate over the tested ghost-angle range may appear to suggest that the spoofing problem is fully solved in the considered setting. However, this interpretation would be misleading. The result should instead be understood in the context of the adopted post-processing formulation, where the transformation is applied directly to recorded IQ data rather than being realized through a full over-the-air multi-antenna transmission.

Under these conditions, the spoofing transformation operates in a particularly controlled setting and it is a consequence of the algebraic exactness of the transformation under ideal conditions. With a two-element receiver ($N = 2$) and a single dominant source, the measured signal can be projected onto the baseline steering direction and then reconstructed along the target ghost direction without introducing additional transmission-side uncertainty: the received signal lies entirely in the 1-dimensional subspace spanned by $a(\theta)$, projecting onto this subspace recovers $\hat{s}(t)$ without information loss; reprojecting onto $a(\phi)$ produces a signal that is, by construction, perfectly consistent with a source at $\phi$ and no noise is added in this process.

This is fundamentally different from the attack as described in [1], where imperfect synchronization between $M$ transmit antennas, channel noise, and multipath all degrade the transformation. The success rates reported by Xu et al. (87% at $M = 5$, $> 90\%$ at $M \geq N = 6$) reflect these real-world imperfections. As a consequence, the excellent quantitative performance reflects the algebraic consistency of the method in this simplified scenario more than the practical feasibility of the complete physical-layer attack.

### B. Real-World Non-Idealities

Although the considered variant avoids the complexity of synchronized multi-antenna transmission, the hardware experiments still reveal several non-ideal effects that are important for a realistic interpretation of the results.

**Phase calibration offset.** A first effect is the systematic phase offset introduced by the two independent local oscillators of the UBX daughterboards. Without calibration, this offset would bias the estimated arrival direction independently of the true transmitter position, causing MUSIC to report a consistently wrong angle and making any subsequent evaluation unreliable.

**Multipath.** A second relevant effect is indoor multipath. The measured MUSIC spectra are not perfectly sharp and exhibit both a broad main lobe and secondary components (visible in Figure 4), which are consistent with reflections in the environment and with the limited angular resolution of a two-element array. It also produced run-to-run variability in the wide-angle estimates (Table II). A narrower mainlobe would require more antennas ($N > 2$).

**Antenna ordering.** The experiments also show a consistent sign inversion between nominal and estimated direction in the wide-angle datasets, which can be explained by the physical ordering of the antennas relative to the assumed ULA model. These effects do not invalidate the transformation, but they make clear that real hardware behavior is already more complex than the idealized picture suggested by simulation alone.

### C. Scope of the Present Contribution

For this reason, the contribution of the paper should be framed carefully. The work does not claim a full hardware realization of the original spoofing attack proposed by Xu et al. [1]. Instead, it provides an experimental validation of the mathematical core of the transformation under real RF measurements acquired with commercial SDR equipment. This distinction is essential, because the original attack assumes multiple synchronized transmit antennas and genuine over-the-air signal injection, whereas the present study replaces that stage with an offline manipulation step imposed by hardware constraints.

Seen in this way, the present results are valuable precisely because they isolate one question: whether the transformation itself remains meaningful when confronted with real measurements rather than synthetic simulations. The answer appears to be positive, but only within the boundaries of the adopted setup and assumptions.

### D. Limitations

The main limitations of the current study follow directly from the chosen experimental configuration:

1) the attack is implemented only in post-processing and therefore does not reproduce the full physical signal injection between attacker and receiver;

2) the receiver uses only two antennas ($N = 2$), which inherently limits angular resolution and makes the spectra more sensitive to ambiguity and broad lobes;

3) all measurements are collected indoors, where multipath is significant and difficult to control.

## VIII. Future Work

A natural next step is therefore the transition toward a more faithful hardware realization of the original attack model. This would require synchronized multi-antenna SDR transmitters capable of generating the intended spoofing field over the air, together with a more systematic characterization of calibration stability, channel effects, and environment-dependent performance. Such an extension would make it possible to bridge the current gap between mathematical validation and full physical deployment.

## IX. Conclusion

This work presents a hardware-based validation of the mathematical core of the DOA spoofing attack originally proposed by Xu et al. [1]. Rather than attempting to reproduce the full over-the-air multi-antenna attack, we adapted the transformation to a constrained USRP setup with $M = 1$ transmitted antenna and $N = 2$ receiver antennas, applying it directly to recorded IQ samples. This post-processing variant made it possible to evaluate the spoofing mechanism on real RF data while remaining compatible with the available hardware.

On real RF measurements collected at three approximate transmitter positions, the MUSIC estimator was successfully redirected to any desired ghost angle in $[-60°, +60°]$ with a 100% success rate, validating the mathematical foundation of the attack on real hardware. The obtained results show that the estimator can be redirected consistently toward selected ghost angles across the tested range, with zero error in the considered sweep.

At the same time, the measurements highlight practical non-idealities such as phase calibration offsets, multipath, and angular aliasing beyond $\pm 90°$, confirming that a real physical deployment would require substantially more complex engineering than the simplified proof of concept considered here.

Overall, the paper establishes a reproducible experimental baseline for future synchronized multi-antenna implementations and provides a concrete step toward understanding how the original spoofing idea behaves outside simulation.

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